AI Reference Report

Assignment 02 - Web Mario

1. AI Tool(s) Used

Claude (Anthropic) - claude.ai web interface, used throughout the development process for code generation, debugging, and project architecture guidance.

2. Scope of Usage

The following table lists all files where AI-generated or AI-assisted code was used. All code was reviewed, tested, and modified to fit the project requirements.

File	Lines	Usage Description
Player.ts	All	Player movement, jump, physics collision, hurt/die logic, big/small state switching. Modified multiple times to fix collisi
Enemy.ts	All	Goomba enemy AI: auto patrol, wall bounce, stomp detection. Modified to add turn cooldown and cliff detection.
FlyingGoomba.ts	All	Flying enemy that moves in a sine-wave pattern within a set boundary. Customized fly height and speed parameters.
ChasingGoomba.ts	All	Chasing enemy that detects player and rushes toward them. Added 2-HP system, angry sound effect, and cliff detect
GameManager.ts	All	Singleton game manager for score, lives, coins, timer. Modified to use cc.find for AudioManager lookup.
AudioManager.ts	All	Singleton audio manager for BGM and SFX. Modified to prevent duplicate BGM playback.
PlayerAnimator.ts	All	Sprite frame animation controller for player walk/jump/idle/dead states.
EnemyAnimator.ts	All	Sprite frame animation controller for enemy walk animation.
QuestionBlock.ts	All	Question block interaction: spawn coin or mushroom when hit from below. Modified coin spawn position and audio pla
Mushroom.ts	All	Super mushroom power-up: auto movement, collected by player to grow big.
CameraFollow.ts	All	Camera horizontal follow with smooth lerp and min/max bounds.
ParallaxBackground.ts	All	Parallax scrolling background effect.
UIManager.ts	All	HUD display: score, coins, lives, timer. Reads from GameManager singleton.
MenuScene.ts	All	Start menu with buttons for Start Game, Level Select, and Leaderboard.
LevelSelect.ts	All	Level selection scene with buttons for each level.
WinScene.ts	All	Win screen with score display, name input for leaderboard submission.
GameOverScene.ts	All	Game over screen with score, leaderboard, and retry/menu buttons.
Leaderboard.ts	All	Leaderboard display using localStorage, shows top 5 scores.

3. Prompt and Response Examples

Initial project setup

Prompt: Help me build a Mario-style platformer game using Cocos Creator 2.4.8 with TypeScript. I need basic features: start menu, level select, game view, physics, player control, enemies, question blocks, UI.

Refinement: Claude generated the initial project architecture with 15 TypeScript files covering all basic game systems. I then iteratively refined each script through testing and debugging.

Fixing collision detection

Prompt: Players stomp on Goomba but still die. The onBeginContact normal vector judgment seems wrong.

Refinement: Claude suggested replacing normal-vector-based detection with position-based detection (comparing player bottom Y vs enemy top Y), which resolved the issue.

Audio not playing

Prompt: AudioManager.instance clips are all null after scene reload.

Refinement: Claude identified that addPersistRootNode requires the node to be a direct child of the scene root, not under Canvas. Moving AudioManagerNode outside Canvas fixed the issue.

Adding chasing enemy

Prompt: Create a Goomba that chases the player when detected, rushes in one direction, and needs 2 stomps to kill.

Refinement: Claude generated ChasingGoomba.ts with patrol/chase modes, world-coordinate player detection, and a 2-HP stomp system. I modified it to add angry sound effects and adjusted detection range.

4. Refinement Process

All AI-generated code went through extensive manual testing and modification. Key refinements include:

- Replaced cc.Action (deprecated) with cc.tween for invincibility blinking effect

- Changed AudioManager lookup from static instance to cc.find() to handle cross-scene persistence

- Added world-coordinate conversion for enemy player detection (enemies and player in different node hierarchies)

- Fixed PhysicsBoxCollider sizes to match actual sprite dimensions (was defaulting to 100x100)

- Enabled contact listeners on all RigidBody components (Cocos Creator 2.4.8 requires explicit opt-in)

- Set friction to 0 on player collider to prevent wall-sticking

- Added turn cooldown to enemies to prevent rapid direction flipping

- Modified cliff detection to use ground contact tracking instead of raycasting (raycast coordinates were incorrect in nested node hierarchies)

Note: Screenshots of the full conversation are available upon request. The development process involved continuous back-and-forth iteration with Claude to debug physics, audio, collision, and UI issues specific to the Cocos Creator 2.4.8 environment.